

ANSWER KEY

GATE CS — GATE CS 2026 Trend-Based Mock Paper

4

Q.No.	Type	Answer	Marks
1	3MCQ	A	3
2	3MCQ	21	3
3	3MCQ	D	3
4	2NAT	3500	2
5	2MCQ	C	2
6	2MCQ	C	2
7	2NAT	A	2
8	2MCQ	C	2
9	2MCQ	B	2
10	2MCQ	B	2
11	1NAT	C	1
12	1MCQ	C	1
13	1MCQ	B	1
14	1MCQ	B	1
15	1NAT	B	1
16	1MCQ	B	1
17	1NAT	A	1
18	1MCQ	C	1
19	1MCQ	C	1
20	1MCQ	C	1
21	1MCQ	B	1
22	1NAT	B	1
23	2MCQ	C	2
24	1MCQ	C	1
25	2MSQ	C	2
26	1NAT	B	1
27	1MCQ	B	1
28	1MCQ	A	1
29	1MCQ	B	1
30	1NAT	A	1
31	1NAT	C	1
32	1MCQ	B	1
33	1MCQ	C	1
34	1NAT	B	1
35	1NAT	C	1
36	1MCQ	B	1

37	1NAT	B	1
38	1NAT	A	1
39	1MCQ	B	1
40	1NAT	B	1
41	1MCQ	B	1
42	1NAT	B	1
43	1MCQ	B	1
44	1MCQ	A	1
45	2NAT	10	2
46	2NAT	2	2
47	2MSQ	A	2
48	2MCQ	A	2
49	2MCQ	48	2
50	2MCQ	0.6	2
51	2NAT	4	2
52	2MCQ	435	2
53	2MCQ	D	2
54	2MCQ	A	2
55	2MCQ	A	2
56	2NAT	Register allocation	2
57	2MCQ	A	2
58	1NAT	B	1
59	2MCQ	0	2
60	2NAT	A	2
61	2MCQ	E2	2
62	2MCQ	A	2
63	2MSQ	A	2
64	2MCQ	A	2
65	2NAT	4	2

Q1. General Aptitude | 3 marks | 3MCQ

According to the map shown in the figure, which one of the following statements is correct? Note: The figure shown is re...

Answer: A

From the map: Library is northwest of Canteen.

Source: Adapted from GATE 2025 CS Session 1 Q5

Q2. General Aptitude | 3 marks | 3MCQ

In the sequence 6, 9, 14, x, 30, 41, a possible value of x is

Answer: 21

The successive differences form consecutive odd numbers: $9 - 6 = 3$, $14 - 9 = 5$, so the next differences should be 7, 9, 11. Adding 7 to 14 gives 21, then $21 + 9 = 30$ and $30 + 11 = 41$, matching the given terms. Hence $x = 21$ (option B).

Source: Adapted from GATE 2024 CS Session 2 Q5

Q3. General Aptitude | 3 marks | 3MCQ

Let 1 and 2 denote two arbitrary prime numbers. Which one of the following statements is correct for all values of 1 and...

Answer: D

For primes p , p , $p^2 + p$, $p^2 + 1$ can be prime. Check all options: only (D) is always true for all prime pairs.

Source: Adapted from GATE 2025 CS Session 2 Q5

Q4. General Aptitude | 2 marks | 2NAT

In an engineering college of 10,000 students, 1,500 like neither their core branches nor other branches. The number of S...

Answer: 3500

Total 10000, neither=1500. Using inclusion-exclusion: Core only = 3500.

Source: Adapted from GATE 2024 CS Session 2 Q3

Q5. General Aptitude | 2 marks | 2MCQ

Bird : Nest :: Bee : _____. Select the correct option to complete the analogy.

Answer: C

A bird lives in a nest, just as a bee lives in a hive. Hence the correct pair is Bee : Hive.

Source: Adapted from GATE 2025 CS Session 2 Q2

Q6. General Aptitude | 2 marks | 2MCQ

If ' ' denotes increasing order of intensity, then the meaning of the words [walk jog sprint] is analogous to [bothered ...

Answer: C

Walk < jog < sprint (increasing intensity). Bothered < fazed < daunted.

Source: Adapted from GATE 2024 CS Session 2 Q1

Q7. General Aptitude | 2 marks | 2NAT

In a class of 60 students, 30 play cricket, 25 play football, and 15 play both. How many play neither?

Answer: A

Correct Answer: A

Source: Adapted from GATE 2024 CS Session 1 Q10

Q8. General Aptitude | 2 marks | 2MCQ

Ravi had _____ younger brother who taught at _____ university. He was widely regarded as _____ honorable man. Select ...

Answer: C

Articles: 'a' younger brother, 'an' university (vowel sound), 'an' honorable man (silent h).

Source: Adapted from GATE 2025 CS Session 1 Q1

Q9. General Aptitude | 2 marks | 2MCQ
Antonym of 'ephemeral':

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q10

Q10. Engineering Mathematics | 2 marks | 2MCQ
Determinant of 2x2 identity:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 CS Session 1 Q1

Q11. Engineering Mathematics | 1 marks | 1NAT
Probability of at least one head in 3 coin tosses:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q11

Q12. Engineering Mathematics | 1 marks | 1MCQ
Let A be a 3x3 matrix with eigenvalues 1, 2, and 3. Then the trace of A^2 is:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q1

Q13. Engineering Mathematics | 1 marks | 1MCQ
Derivative of x^3 at $x=2$:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q1

Q14. Engineering Mathematics | 1 marks | 1MCQ
Mean of binomial $n=10, p=0.5$:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 CS Session 1 Q1

Q15. Programming and Data Structures | 1 marks | 1NAT
Infix to postfix uses:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 CS Session 1 Q6

Q16. Programming and Data Structures | 1 marks | 1MCQ
What is the time complexity of inserting at the beginning of a linked list?

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q3

Q17. Programming and Data Structures | 1 marks | 1NAT
Inorder traversal of BST gives:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 CS Session 1 Q3

Q18. Programming and Data Structures | 1 marks | 1MCQ
Avg hash lookup:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2021 CS Session 1 Q3

Q19. Algorithms | 1 marks | 1MCQ
Kruskal uses:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2022 CS Session 1 Q2

Q20. Algorithms | 1 marks | 1MCQ
Merge sort space:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2021 CS Session 1 Q7

Q21. Algorithms | 1 marks | 1MCQ
Time complexity of binary search:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q11

Q22. Algorithms | 1 marks | 1NAT
BFS uses:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 CS Session 1 Q7

Q23. Algorithms | 2 marks | 2MCQ
The time complexity of Dijkstra's algorithm using a binary heap is:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q7

Q24. Algorithms | 1 marks | 1MCQ
LCS uses:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2023 CS Session 1 Q7

Q25. Operating Systems | 2 marks | 2MSQ
Using 3 frames, which page replacement algorithm would not cause any page fault for reference string 0,1,2,3,0,1,4,0,1,2...

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q8

Q26. Operating Systems | 1 marks | 1NAT
Semaphore with value 1 is:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q8

Q27. Operating Systems | 1 marks | 1MCQ
Banker's algorithm is for:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 CS Session 1 Q4

Q28. Operating Systems | 1 marks | 1MCQ
Round-robin uses:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 CS Session 1 Q4

Q29. Operating Systems | 1 marks | 1MCQ
Inode contains:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 CS Session 1 Q4

Q30. Operating Systems | 1 marks | 1NAT
TLB stands for:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2022 CS Session 1 Q9

Q31. Operating Systems | 1 marks | 1NAT
Which of the following is NOT a necessary condition for deadlock?

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q4

Q32. Computer Networks | 1 marks | 1MCQ
Protocol to find MAC from IP:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q6

Q33. Computer Networks | 1 marks | 1MCQ
Which protocol operates at the application layer of the TCP/IP model?

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q6

Q34. Computer Networks | 1 marks | 1NAT
CSMA/CD is used in:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q13

Q35. Computer Networks | 1 marks | 1NAT
Asymmetric encryption:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2022 CS Session 1 Q5

Q36. Computer Networks | 1 marks | 1MCQ
OSPF is:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 CS Session 1 Q6

Q37. Computer Networks | 1 marks | 1NAT
TCP connection establishment uses:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q14

Q38. Digital Logic | 1 marks | 1NAT
Flip-flop stores:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2021 CS Session 1 Q2

Q39. Digital Logic | 1 marks | 1MCQ
The number of Boolean functions of n variables is:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q2

Q40. Digital Logic | 1 marks | 1NAT
Full adder has how many inputs:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q2

Q41. Theory of Computation | 1 marks | 1MCQ
Halting problem:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 CS Session 1 Q8

Q42. Theory of Computation | 1 marks | 1NAT
PDA recognizes:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q14

Q43. Theory of Computation | 1 marks | 1MCQ
Which language is regular?

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q13

Q44. Compiler Design | 1 marks | 1MCQ
Lexical analyzer outputs:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 CS Session 1 Q15

Q45. CS | 2 marks | 2NAT
The pseudocode of a function `fun()` is given below: `fun(int A[0,...,n-1]){ for i=0 to n-2 for j=0 to n-i-2 if (A[j]>A[j+1]...`

Answer: 10

Bubble sort pseudocode. For n elements, inner loop runs $n(n-1)/2$ times = 10 comparisons for $n=5$.

Source: Adapted from GATE 2025 CS Session 1 Q33

Q46. CS | 2 marks | 2NAT
Which of the following fields is/are modified in the IP header of a packet going out of a network address translation (NAT)...

Answer: 2

NAT modifies source IP and source port in IP header. Two fields total.

Source: Adapted from GATE 2024 CS Session 1 Q31

Q47. CS | 2 marks | 2MSQ
Consider the following grammar, with s as the start symbol. The grammar has three incomplete productions denoted by (1), ...

Answer: A

Grammar productions for balanced parentheses. Correct completions form valid CFG.

Source: Adapted from GATE 2024 CS Session 1 Q38

Q48. CS | 2 marks | 2MCQ
two semaphores: s_1 (initialized to 1) and s_2 (initialized to 0). The threads also share a global variable x (initialized...

Answer: A

T1 acquires s_1 , then blocks on s_2 . T2 can't acquire s_1 . If T1 runs first: prints 1, then T2 prints 2.

Source: Adapted from GATE 2024 CS Session 2 Q46

Q49. CS | 2 marks | 2MCQ

The value of the definite integral $\int_{-3}^{-2} \int_{-1}^0 \int_0^1 (4x^2y - z^3) dz dy dx$ is . (Rounded off to the nearest integer) CS GATE 2022...

Answer: 48

Triple integral $\int_{-3}^{-2} \int_{-1}^0 \int_0^1 (4x^2y - z^3) dz dy dx$ over limits. Computing: 48.

Source: Adapted from GATE 2023 CS Session 1 Q31

Q50. CS | 2 marks | 2MCQ

Consider a demand paging system with four page frames (initially empty) and LRU page replacement policy. For the followi...

Answer: 0.6

Simulating LRU with 4 frames over the 15 references yields 9 page faults. Fault rate = $9/15 = 0.6$ (rounded to one decimal).

Source: Adapted from GATE 2022 CS Session 1 Q64

Q51. CS | 2 marks | 2NAT

Consider the following context-free grammar , where , , and are the variables (non-terminals), and are the terminal symb...

Answer: 4

CFG generates strings of length 4. Counting using grammar rules: 4 strings.

Source: Adapted from GATE 2025 CS Session 1 Q19

Q52. CS | 2 marks | 2MCQ

The pseudocode of a function fun() is given below: fun(int A[0,...,n-1]){ for i=0 to n-2 for j=0 to n-i-2 if (A[j]>A[j+1]...

Answer: 435

Bubble sort performs a swap for each inversion; a descending array of 30 elements has $30 \times 29/2 = 435$ inversions.

Source: Adapted from GATE 2025 CS Session 1 Q33

Q53. CS | 2 marks | 2MCQ

(.) is a function from to , (.) is a function from to , and their composition defined as ((.)) is a mapping from to

Answer: D

Even if g and g" f are onto, f need not be onto or one to one; only g must be onto.

Source: Adapted from GATE 2025 CS Session 1 Q17

Q54. CS | 2 marks | 2MCQ

Which of the following statements about a relation R in first normal form (1NF) is/are TRUE ?

Answer: A

1NF only requires atomic attributes; a relation may have a composite key, foreign keys, and multiple candidate keys.

Source: Adapted from GATE 2024 CS Session 1 Q22

Q55. CS | 2 marks | 2MCQ

Consider a binary min-heap containing 105 distinct elements. Let be the index (in the underlying array) of the maximum e...

Answer: A

In a min heap the maximum element must be at a leaf. For 105 nodes, leaves are indices 53 to 105, i.e., 53 possible positions.

Source: Adapted from GATE 2024 CS Session 1 Q43

Q56. CS | 2 marks | 2NAT

Which ONE of the following techniques used in compiler code optimization uses live variable analysis?

Answer: Register allocation

Live variable analysis is used to determine variable lifetimes, which drives register allocation.

Source: Adapted from GATE 2025 CS Session 1 Q13

Q57. CS | 2 marks | 2MCQ

In a B+ tree, the requirement of at least half-full (50%) node occupancy is relaxed for which one of the following cases...

Answer: A

The 50% occupancy rule is relaxed only for the root node in a B+ tree.

Source: Adapted from GATE 2024 CS Session 1 Q21

Q58. General Aptitude | 1 marks | 1NAT

Synonym of 'ubiquitous':

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 CS Session 1 Q10

Q59. CS | 2 marks | 2MCQ

Let A be the adjacency matrix of the graph with vertices $\{1, 2, 3, 4, 5\}$. Let 1, 2, 3, 4, and 5 be the five eig...

Answer: 0

For any square matrix, the sum of its eigenvalues (counting multiplicities) equals the trace of the matrix. An adjacency matrix of a simple graph has zeros on its diagonal (no self loops), so its trace is 0. Hence the sum of the five eigenvalues of A is 0.

Source: Adapted from GATE 2023 CS Session 1 Q30

Q60. CS | 2 marks | 2NAT

Consider the following statements about the use of backpatching in a compiler for intermediate code generation: (I) Back...

Answer: A

Backpatching can handle both Boolean expressions and flow control statements in a single pass.

Source: Adapted from GATE 2025 CS Session 2 Q21

Q61. CS | 2 marks | 2MCQ

A packet with the destination IP address 145.36.109.70 arrives at a router whose routing table is shown. Which interface...

Answer: E2

IP 145.36.109.70 matches E2 entry in routing table (longest prefix match).

Source: Adapted from GATE 2025 CS Session 1 Q40

Q62. CS | 2 marks | 2MCQ

Consider four processes $P, Q, R,$ and S scheduled on a CPU as per round robin algorithm with a time quantum of 4 units. T...

Answer: A

Round robin with quantum 4: $P(0,8)!4, Q(1,5)!10, R(2,3)!6, S(3,2)!2$. Completion times match option A.

Source: Adapted from GATE 2022 CS Session 1 Q42

Q63. CS | 2 marks | 2MSQ

`#include <stdio.h> int foo(int S[], int size){ if(size == 0) return 0; if(size == 1) return 1; if(S[0] != S[1]) return 1+...`

Answer: A

Recursive foo counts distinct adjacent pairs. For appropriate input, returns value matching option A.

Source: Adapted from GATE 2025 CS Session 1 Q61

Q64. CS | 2 marks | 2MCQ

Which one of the following CIDR prefixes exactly represents the range of IP addresses 10.12.2.0 to 10.12.3.255?

Answer: A

10.12.2.0 to 10.12.3.255 spans two /24 networks. /23 prefix 10.12.2.0/23 exactly covers this range.

Source: Adapted from GATE 2024 CS Session 2 Q38

Q65. CS | 2 marks | 2NAT

Consider the following C function definition. `int f(int x, int y) { for (int i=0; i<y; i++) { x=x+x+y; } return x; }` Whi...

Answer: 4

f(2,3): i=0: x=2+2+3=7, i=1: x=7+7+3=17, i=2: x=17+17+3=37. Number of additions = 4.

Source: Adapted from GATE 2024 CS Session 1 Q48
